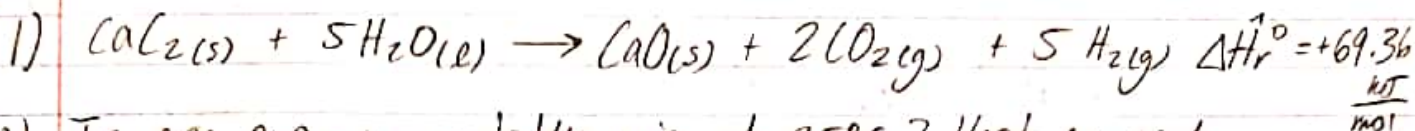


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Sarah Pabilic

CHE 210 HW #10



- a) Is rxn exo- or endothermic at 25°C? Heat or cool reactor to keep T constant? What would T do if reactor ran adiabatically? What can you infer about the energy required to break the molecular bonds of the reactants and that released when product bonds are formed?

Reaction is endothermic so the reactor needs to be heated up to keep T constant. If the reactor is adiabatic, $Q=0$. However, the temperature could still change. Since the reaction is endothermic, the temperature would decrease. Additionally, with the reaction being endothermic, it can be inferred that the energy needed to break the bonds is greater than the energy released when forming bonds.

- b) Calculate ΔU_r° for this reaction. Briefly explain the physical significance of your calculated value.

Since this reaction takes place in the liquid/solid phase, $\Delta U_r^\circ = \Delta H_r^\circ = +69.36 \text{ kJ/mol}$. Because this value is positive, it means that the energy goes from the surroundings to the system.

- c) Suppose you charge 150.0g of CaC_2 and liquid water into a rigid container at 25°C, heat the container until the calcium carbide reacts completely, and cool the pots back down to 25°C, condensing all the unconsumed water. Write and simplify the energy balance eq. for this closed constant-volume system & use it to determine the net amount of heat (kJ) that must be transferred to

(or from) the reactor. Indicate direction of heat transfer

$$\Delta U + \cancel{\Delta E_k} + \cancel{\Delta E_p} = Q + \cancel{W}$$

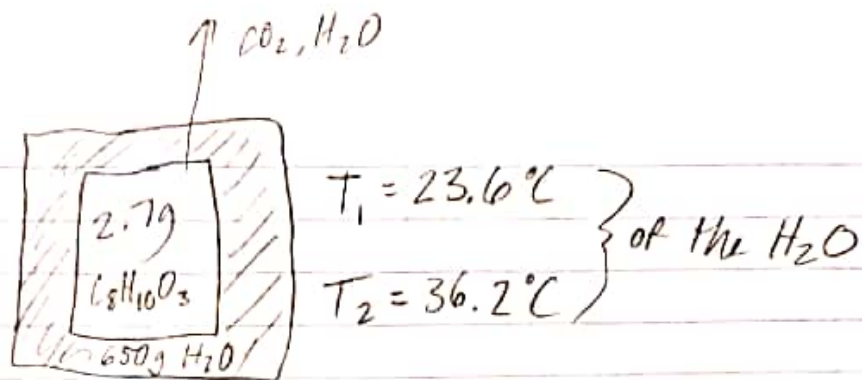
$$\Delta U = Q$$

$$\Delta \hat{U}_r = Q$$

$$1.352 \text{ mol CaCl}_2 (69.36 \text{ kJ/mol}) = Q$$

$Q = 93.77 \text{ kJ}$, positive so heat flows from surroundings to system

2)
(6.32)



Calculate enthalpy of formation (kJ/g) for the unknown chemical



$$\Delta U + \cancel{\Delta E_k} + \cancel{\Delta E_p} = Q + \cancel{W}$$

$$\Delta U_r = Q_{\text{sys}} \quad Q = -mC\Delta T = 0.65 \text{ kg} (4.19 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}) (36.2 - 23.6 ^\circ\text{C})$$

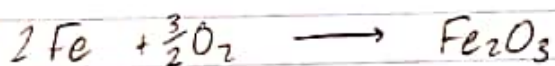
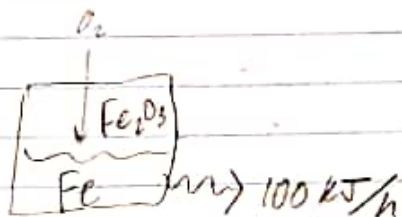
$$Q_{\text{sys}} = -34.3 \text{ kJ (sys gives off heat)}$$

$$(2.7\text{g})H_{f, \text{C}_8\text{H}_{10}\text{O}_3} + (-9)(0) + (8)(-393.5 \frac{\text{kJ}}{\text{g mol}}) + (5)(-285.84 \frac{\text{kJ}}{\text{g mol}}) = -34.3$$

$$H_{f, \text{C}_8\text{H}_{10}\text{O}_3} = 1682.56 \frac{\text{kJ}}{\text{g}}$$



- 3) Hotpacks; Oxygen reacts with iron, producing iron oxides, mainly Fe_2O_3 . Consumers want a slow release of about 100 kJ/h and last about 6 h . How much Fe would you put in the hot pack? How much would the mass of the hotpack change over the course of its use (total g and %)?



$$\dot{\Delta H} + \dot{\Delta E}_K + \dot{\Delta E}_P = \dot{Q} + \dot{W}$$

$$m(-5.201 \frac{\text{kJ}}{\text{g}}) - (2)(0) - \frac{3}{2}(0) = -100 \frac{\text{kJ}}{\text{h}}$$

$$m = 19.23 \frac{\text{g}}{\text{h}} \times 6 \text{ h} \Rightarrow m_f = 115.36 \text{ g Fe}_2\text{O}_3$$

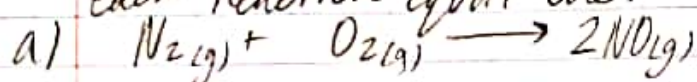
115.36 g Fe_2O_3	1 mol	2 mol Fe	55.845 g
	159.69 g	1 mol Fe_2O_3	1 mol Fe

80.68 g Fe initially in pack

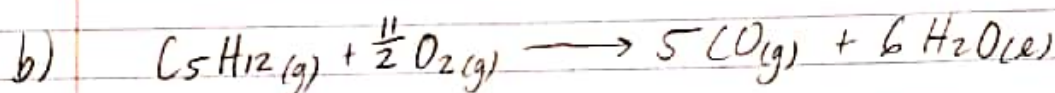
$$115.36 \text{ g} - 80.68 \text{ g} = 34.68 \text{ g increase}$$

$$\% \text{ increase} = \frac{34.68 \text{ g}}{80.68 \text{ g}} = 0.4298 \Rightarrow 42.98 \%$$

4) Use tabulated heats of formation to determine the standard heats of the following rxns in kJ/mol, letting the stoichiometric coefficient of the first reactant in each reaction equal one.

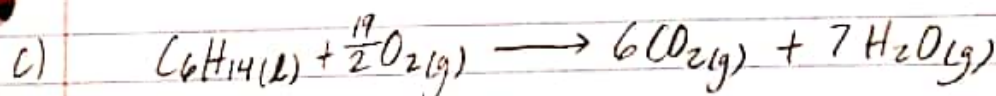


$$\Delta \hat{H}_r^\circ = (2)(90.25 \frac{\text{kJ}}{\text{mol}}) - 0 - 0 \Rightarrow \Delta \hat{H}_r^\circ = 180.5 \frac{\text{kJ}}{\text{mol}}$$



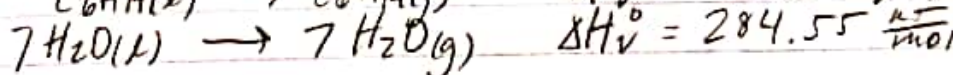
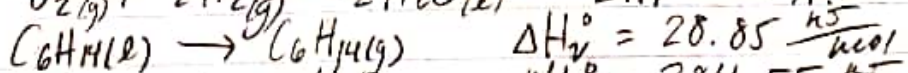
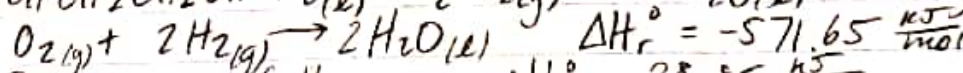
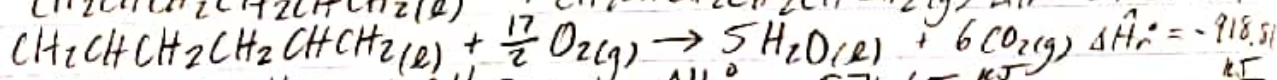
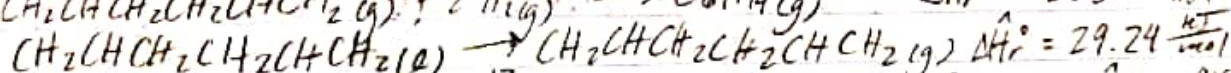
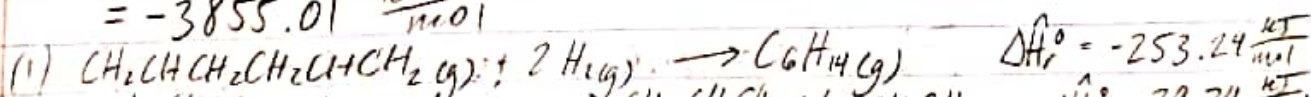
$$\Delta \hat{H}_r^\circ = 5(-110.53) + 6(-285.84) - (-146.76) - \frac{11}{2}(0)$$

$$\Delta \hat{H}_r^\circ = -2120.93 \frac{\text{kJ}}{\text{mol}}$$



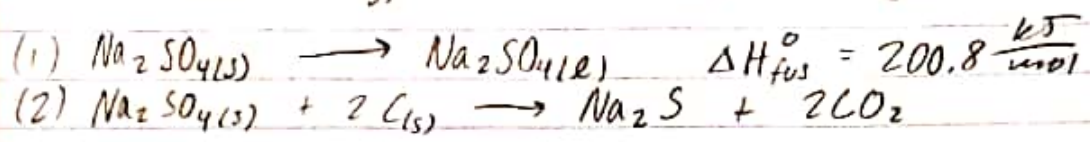
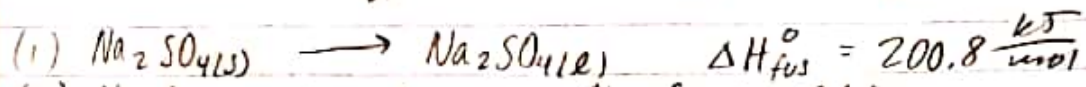
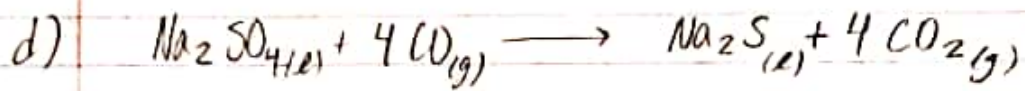
$$\Delta \hat{H}_r^\circ = 7(-241.83) + 6(-393.5) - (-148.8) - \frac{19}{2}(0)$$

$$= -3855.01 \frac{\text{kJ}}{\text{mol}}$$

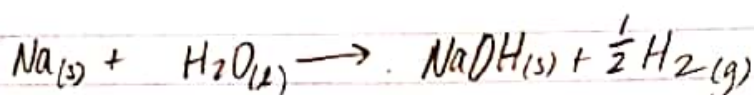


$$\Delta \hat{H}_r^\circ = (-1)(-253.24) + (-1)(29.24) + (-918.51) + (-571.65) + 28.85 + 284.55$$

$$= -953.06 \frac{\text{kJ}}{\text{mol}}$$



5) (6.37) A pure sodium crystal (100g) sits in a steel cup. One drop (1 mL) water is added. H_2 gas is produced. What else happens? Explain, as quantitatively as possible, why it's dangerous to add water to pure metals like Na.



$$\Delta \hat{H}_r^\circ = (-425.9) - (-285.84) = -140.06 \frac{kJ}{mol}$$

$$1 \text{ mL } H_2O = \frac{1 \text{ g } H_2O}{18.02 \text{ g}} \times \frac{1 \text{ mol}}{1 \text{ mol } H_2O} \times \frac{1 \text{ mol Na}}{22.99 \text{ g}} = 1.276 \text{ g Na react}$$

$$100 \text{ g} - 1.276 \text{ g} = 98.7 \text{ g remain}$$

$$\Delta \hat{H}_r^\circ = -Q$$

$$-140.06 \frac{kJ}{mol} (0.055 \text{ mol } H_2O) = -mC\Delta T$$

$$= -98.7 \text{ g} (0.00123 \frac{kJ}{g^\circ C}) \Delta T$$

$$\Delta T = 63.45^\circ C$$

The large temperature change means that there is enough energy being outputted that the hydrogen ignites

Comment Summary

Page 1

1. -3, incorrect value

Page 3

2. -7 incorrect value

Page 5

3. -3 incorrect value